TBA3241 Social Media Analytics

AY25/26 Semester 2

Final Report

Group 1

HAN LIHUI (A0265957J)

LI XUAN (A0287600E)

JERIN PAUL (A0227170M)

ZHANG LONGSHENG (A0287626N)

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Executive Summary

<to be added after analysis is done>

Data & Network Construction

Data Source

This study uses the Twitter Interaction Network for the U.S. Congress, obtained from the Stanford Network Analysis Project (SNAP). The dataset consists of publicly available Twitter interactions among members of the 117th U.S. Congress.

The data was originally collected using Twitter's API, which the researchers used the Tweepy Python package to interact with, retrieving the most recent 3,200 tweets (the limit set by the API) as of data collection date on 9 June 2022 (Fink, Omodt, Zinnecker, & Sprint, 2023).

Data Cleaning

From the above 3,200 tweets, some data cleaning was done by the researchers. Tweets that were created outside of a common time window (9 February 2022 to 9 June 2022) were excluded. Tweets made by Congressional members who tweeted less than 100 times during the same time window were also excluded (Fink, Omodt, Zinnecker, & Sprint, 2023). The final dataset contains 475 nodes and 10,222 edges after analysing the interactions amongst the remaining tweets.

Due to the above data cleaning, there are two types of sampling bias present in the dataset. The first type of bias is activity bias, where highly active users are over-represented since tweets made by less active users were excluded. The second type of bias is temporal sampling bias, where the dataset only captures a fixed time window and not long term behaviour.

Network Construction

The network was constructed through the following pipeline. First, tweets and its data were extracted using Twitter API. After data cleaning, interactions between users were identified. These interactions include retweets, replies, mentions, and quote tweets. They are then aggregated to obtain the total number of interactions between each pair of users. These interaction counts were normalised by the total number of tweets produced by source user to derive influence probabilities.

The network is a directed, weighted, and unimodal network, stored as a weighted directed edge list. The nodes represent individual Congress member's Twitter account, and the directed edges represent the interaction initiated by target node to source node, where interaction refers to be retweets, quote tweets, replies, and mentions. The edge weights represent the probability of influence. This is calculated using the ratio of number of interactions to the total number of tweets issued by source node.

Datasets

Congressional Twitter network

The Twitter interaction network for the 117th United States Congress House of Representatives was constructed by first obtaining members' official Twitter handles from `https://pressgallery.house.gov/member-data/members-official-twitter-handles`. The Twitter API was then used to obtain all Tweets by members of Congress between February 9, 2022, and June 9, 2022. (The latter date corresponds to when the data were requested, and the earlier date was the farthest point backward in time at which one user's number of Tweets exceeded the 3200-Tweet limit set by the API.) Only the 475 representatives (out of 535) who issued 100 or more tweets during this time frame were included in the network.

A weighted, directed network was constructed such that the connection from user j to user i was given by

$$\mathcal{P}_{ij} = \frac{n_{ij}^{\text{retweet}} + n_{ij}^{\text{quote}} + n_{ij}^{\text{reply}} + n_{ij}^{\text{mention}}}{N_j^{\text{tweets}}}, \qquad (7)$$

where n_{ij}^{retweet} is the number of times user i retweeted user j, n_{ij}^{quote} is the number of times user i quote tweeted user j, n_{ij}^{reply} is the number of times user i replied to user j, n_{ij}^{mention} is the number of times user i mentioned user j, and N_j^{tweets} is the total number of Tweets issued by user j. The weighted connection, $\mathcal{P}_{ij}$, can therefore be interpreted as the probability that user i is observably influenced by user j.

Figure 1. Excerpt from research paper describing calculation used for edge weightage

The original dataset provided was in an edgelist file format. A conversion to a GEXF file format was done on our side to make it compatible with Gephi software. We used NetworkX library in Python to read the edgelist file and construct the graph, before writing it into a GEXF file.

Network Analysis

<with necessary visualisation presented>

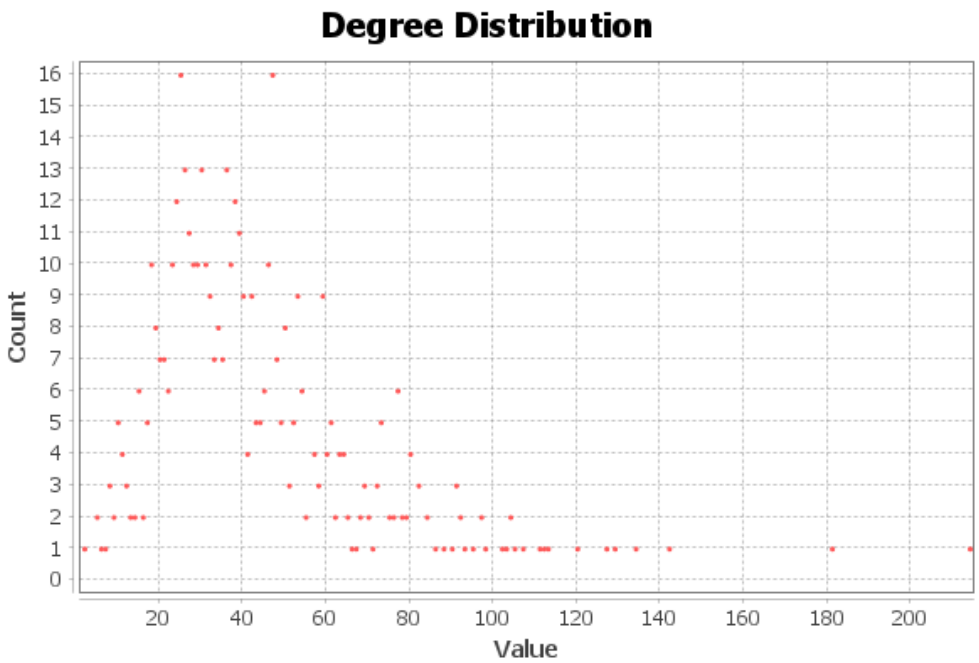
Network Structure

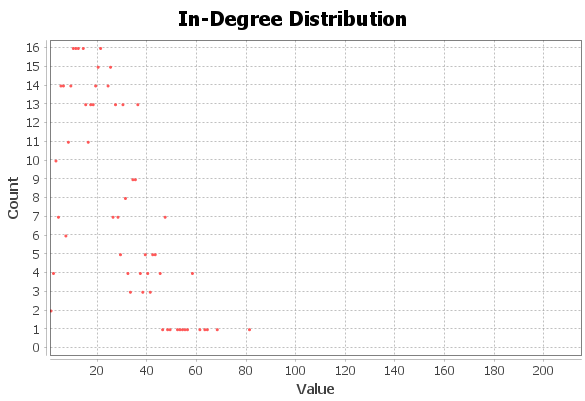
<descriptive network structure, e.g. size, density, diameter>

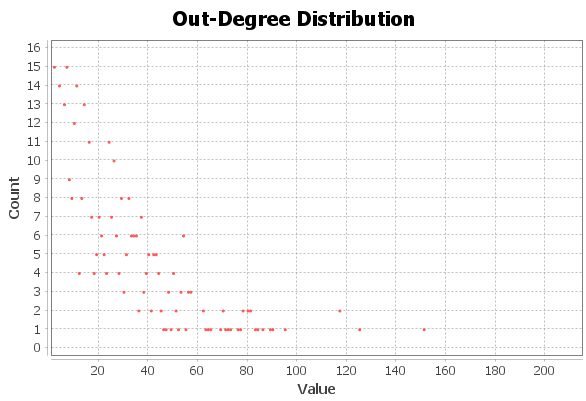
<node level attributes, e.g. centrality measures>

<community structure: community (e.g. cluster) detection and interpretation>

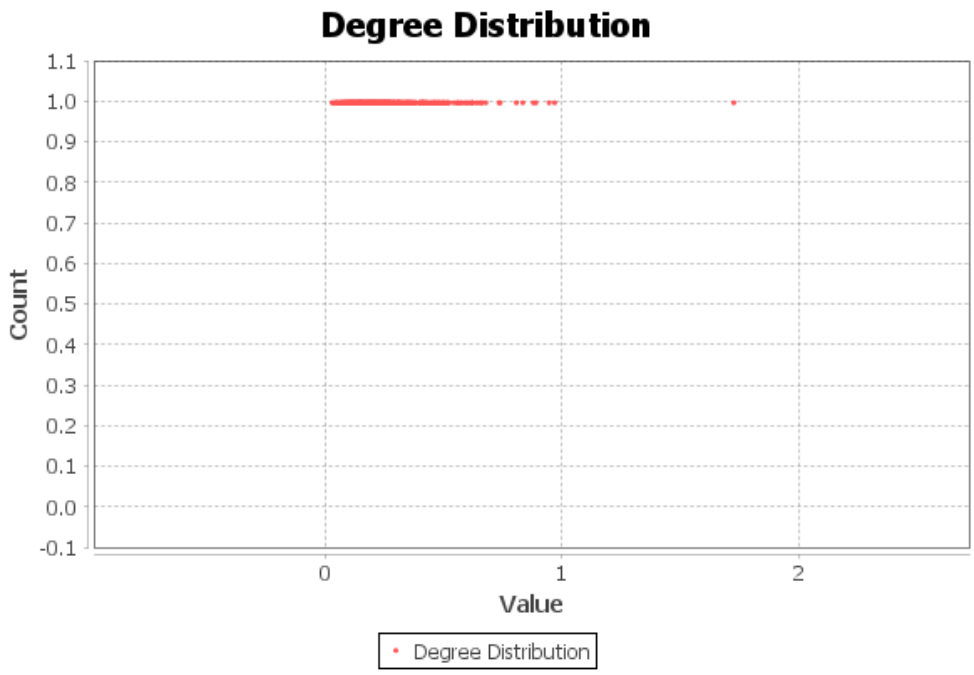
Average Degree: 21.520







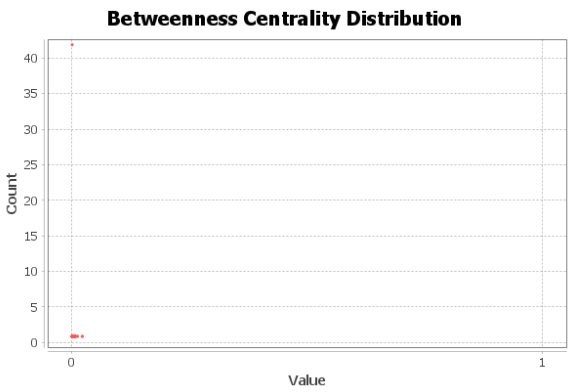
Average Weighted Degree: 0.125

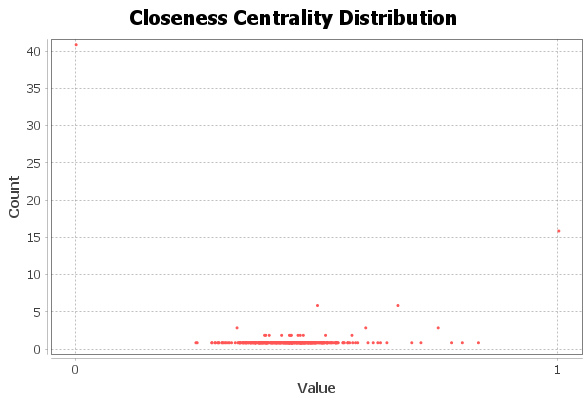


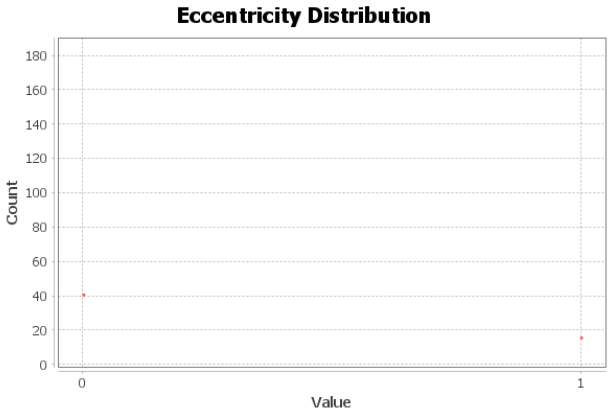
Diameter: 8

Radius: 0

Average Path length: 2.27123479550534







Graph Density: 0.045

Number of Weakly Connected Components: 1

Number of Strongly Connected Components: 475

Average Clustering Coefficient: 0.151 (The Average Clustering Coefficient is the mean value of individual coefficients)

<provide other statistics>

<choose one egocentric network to examine and analyse its structure and dynamics>

Research Questions

Community Differences

<research qn: community differences

- How can the nodes be categorised?
- How do the categories differ from one another?
- Is the Congress Twitter network structurally centralised around a few dominant members?
- Does the network exhibit high degree centralisation or betweenness centralisation?
- Can compare democratic vs republic, senate vs house>

Focus on Influence

<research qn: focus on influence (By identifying the "influential" users in this network, you can also further analyse whether the network level influence is aligned with their real role status in the congress in reality. There might be some interesting findings.)

- Who are the most influential users, and why?
- Is there an imbalance of influence where influence is not reciprocated?
- Are the influential users balanced across parties?
- Can use statistics we learnt in class to examine the centrality and network structure>

Cross Party Influence

<research qn: cross party influence

- To what extent are there cross party influence?
- Are there any critical bridging nodes that act as articulation point?
- Is there a stronger influence within the same party compared to cross party?
- Can potentially evaluate within-group influence vs cross party influence>

What-if Questions

<what if qn: removal of nodes

- How will the overall network cohesion (e.g. number of components, diameter, fragmentation) change if the most influential nodes are removed?
- What if the bridging nodes are removed? Does the network rely on a small subset of nodes to maintain connectivity?>

<what if qn: changes to edge weight

- What if the transmission probabilities are increased or decreased in different scenarios? (LH: imo weak qn) How will this change the graph analysis and the viral centrality of the nodes? (LH: can explore but I don't have much hope)>

Implications

<what do the findings imply in the real world context>

(from proposal: We can potentially extend our analysis to better understand the larger influence structure of the Congress in reality, and apply the same questions in political analyses, such as finding out who exerts most influence, the cluster strength, and structural fragility. Political parties could also potentially use the findings to strengthen intra- and inter-party influence (e.g., who is working as the spokesman on the network for each party?), as well as optimising strategies during election campaigns.)

<identify at least 2 actionable insights, e.g. platform policy, marketing, platform governance, community management>

Conclusion and Limitation

<summarise key findings>

<discuss the limitations in this work>

Reproducibility

The analysis was conducted using Python and Gephi. In Python, the NetworkX library was used to convert the edgelist file provided by SNAP, to a Gephi compatible GEXF file.

```python
import networkx as nx

# Read the edgelist file (default delimiter is whitespace)
# The `read_edgelist` function is available in the NetworkX documentation.
G = nx.read_edgelist('congress.edgelist')

nx.write_gexf(G, 'congress.gexf')
```

The script reads from edgelist file, constructs it as a graph, and writes the graph into a GEXF file.

```xml
<edges>
  <edge source="0" target="4" id="0" weight="0.0036496350364963502" />
  <edge source="0" target="12" id="1" weight="0.002105263157894737" />
  <edge source="0" target="18" id="2" weight="0.002347417840375587" />
  <edge source="0" target="25" id="3" weight="0.004210526315789474" />
  <edge source="0" target="30" id="4" weight="0.002105263157894737" />
```

The resulting GEXF file retains edge information like direction and weight. The GEXF file is then imported into Gephi for network visualisation and analysis. Within Gephi, layout algorithms can be applied to better represent the structure of the network.

References

Fink, C. G., Omodt, N., Zinnecker, S., & Sprint, G. (2023). A Congressional Twitter network dataset quantifying pairwise probability of influence. *Elsevier*.

Fink, C., Fullin, K., Gutierrez, G., Omodt, N., Zinnecker, S., Sprint, G., & McCulloch, S. (2023). A centrality measure for quantifying spread on weighted, directed networks. *Physica A*.

Stanford University. (n.d.). *SNAP: Network datasets: Twitter Interaction Network for the US Congress*. Retrieved from SNAP: https://snap.stanford.edu/data/congress-twitter.html

Declare of AI tool usage

AI Tool used	How is the tool used in the work

Appendix A

<if any>